EAST WEST UNIVERSITY

Physics Lab

Department of MPS

Course Name : Engineering Physics -I

Course Code : PHY 109 LAB

Section No : 08

Group No : 05

Experiment No : 02

Name of the Experiment : To determine the refractive index of the material of a prism.

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PHY-109 LAB

Experiment No: 02

Name of the Experiment: To determine the refraction index of the material of a prism.

Theory: When light of certain colour undergoes by refraction through a prism, then the refractive index of the material of the prism for light of certain colour (wave length) is given by the relation

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin \frac{\delta_m + \angle A}{2}}{\sin \frac{\angle A}{2}} \quad \text{------} (1)$$

Where, $\angle A$ = prism angle
δ = minimum deviation

the expression for μ can be deduced in the following manner:

If a ray PQ (Fig-1) is incident on the first face of a prism and after passing through the principal plane of the prism, finally emerge out through the other face in the direction RS.

Let i and r be the respective angles of incidence and refraction at the first face of the prism and r' and e the corresponding quantities for second face.

Now the deviation (δ) of the ray is

$$\angle STX = \delta = i + e - (r + r') \quad \text{.......} (2)$$

In quadrilateral AQFR,

$$\angle Q = \angle R = 90^\circ \quad [\text{MF and NF are normal}]$$

$$\therefore \angle A + \angle F = 180^\circ \quad \text{.......} (3)$$

Again, in triangle QFR,

$$\angle F + r + r' = 180^\circ \quad \text{———} (4)$$

$$\text{.......} (5)$$

$$\therefore \angle A = r + r' \quad \text{.......} (6)$$

$$\therefore \delta = i + e - \angle A$$

But in the position of minimum deviation (δ_m) the ray passes symmetrically through the prism so that $i = e$ and $r = r'$.

Therefore, we can write fo from (5) and (6),

$$\delta_m = 2i - \angle A \quad \text{and} \quad \angle A = 2r$$

$$\therefore i = \frac{\delta_m + \angle A}{2} \quad \text{and} \quad r = \frac{\angle A}{2}$$

Now refractive index of the ~~medium~~ material of the prism is

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin \frac{\delta_m + \angle A}{2}}{\sin \frac{\angle A}{2}} \quad \ldots\ldots (7)$$

Apparatus : 1. Spectrometer 2. Sodium lamp 3. Prism 4. Spirit Level 5. Reading lens etc.

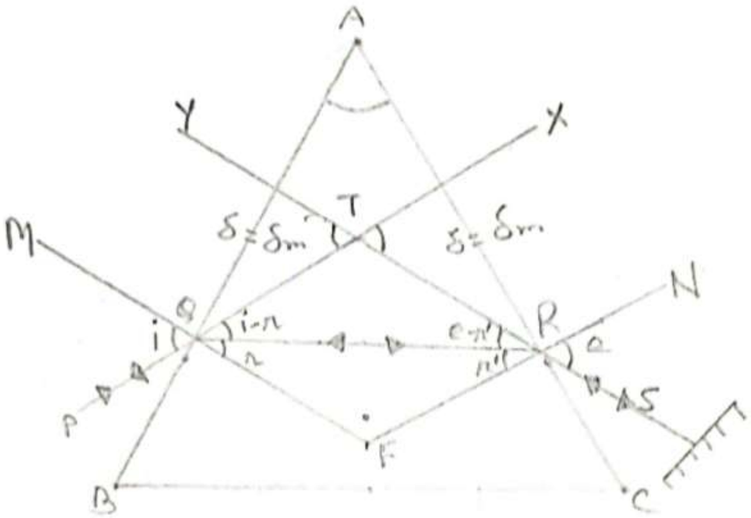

Figure - 1: Refraction of light through prism.

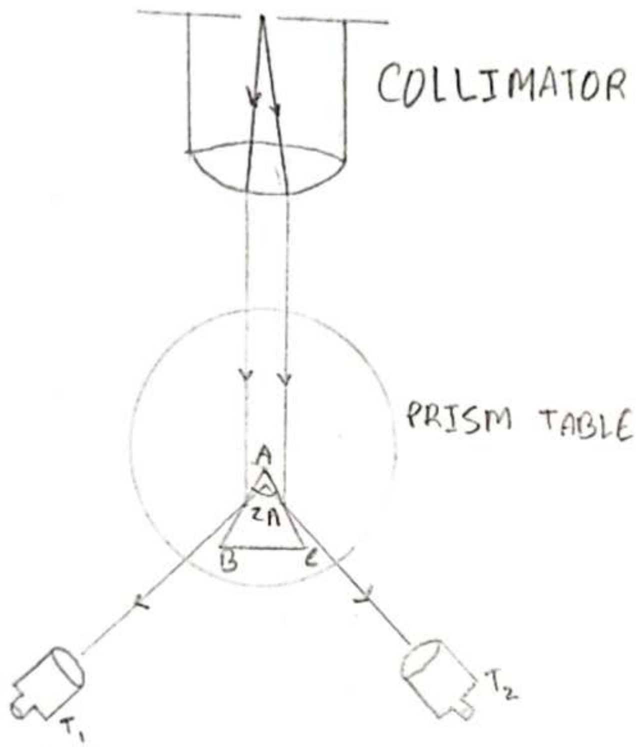

Figure-2: Determination Prism angle

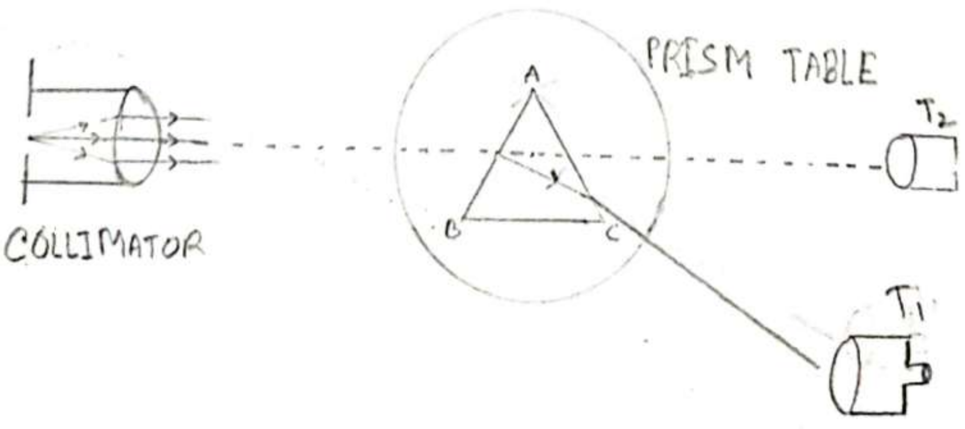

Figure-3: Determination of minimum deviation

DATA SHEET:

$$\text{Vernier ConstAnt (V.C.)} = \frac{\text{Value of the smallest division in Main scale}}{\text{Total number of divisions in Vernier scale}} = 1/60$$

(A) Table for **angle of the prism(A):**

Vernier Scale No	No. of Observation	Reading for image at the Face AB of the Prism: Main Scale Reading (M)	Vernier Scale Reading V = Div. × V.C.	Total Reading T_1=M+V degree	Reading for image at the Face AC of the Prism: Main Scale Reading (M)	Vernier Scale Reading V = Div. × V.C.	Total Reading T_2=M+V degree	$\Theta = T_2 \sim T_1$	Mean Θ degree	Angle of the prism $A = \Theta/2$ degree
I	1	208	0.33	208.33	89	0.45	89.45	118.88	113.865	56.93
II	1	76	0.4	76.4	185	0.25	185.25	108.25		

(B) Table for the **minimum deviation (δ_m):**

Vernier Scale No.	No. of Observation	Reading for **Direct Position**: Main Scale Reading (M)	Vernier Scale Reading V = Div. × V.C.	Total Reading T_1=M+V degree	Reading for **Minimum Deviated** position: Main Scale Reading (M)	Vernier Scale Reading V = Div. × V.C.	Total Reading T_2=M+V degree	$\delta_m = T_2 \sim T_1$ degree	Mean δ_m degree
I	1	126.5	0.2	126.7	91	0.15	91.15	35.55	33.383
II	1	184	0.283	184.283	153	0.067	153.067	31.216	

Calculation:

$$\mu = \frac{Sin\,i}{Sin\,r} = \frac{Sin\,(\frac{\delta_m + \angle A}{2})}{Sin\,(\frac{\angle A}{2})} = \frac{\sin\left(\frac{33.383+56.93}{2}\right)}{\sin\left(\frac{56.93}{2}\right)} = 1.4876$$

$$\text{Percentage Difference} = \frac{\text{Standard value} \sim \text{Experimental value}}{\text{Standard value}} \times 100\%$$

Experimental value of refractive index (measured by experiment) = 1.4876

Standard value of Refractive index = **1.52** (known crown glass)

Percentage difference = $\frac{1.52 - 1.4876}{1.5} \times 100 = 2.13\%$

Results:

Refractive index of material of the prism = 1.4876

Discussions:

(i) The source must be in front of the collimator slit so that image appears bright.

(ii) The vertical cross-wire or the centre of the cross-wires should be made coincident with the same edge of the slit image. Care should be taken so that there is no parallax between the cross-wires and the slit image.

(iii)For the final setting, the telescope must be rotated carefully with the tangent screw always in the same direction so as to avoid back-lash error.

(iv)While measuring the angle of the prism, the vertex of the prism should be placed at the center of the table.

(v) Always use both the verniers to avoid *eccentric error.*

(vi)While handling a prism, never tough the refracting faces. Always hold it between the thumb at the bottom and the other fingers at the top. The faces of the prism should be cleaned with a clean piece of fine linen, if necessary.

(vii) The width of the slit image should be as narrow as possible.

(viii) In taking reading care should be taken to ascertain whether the zero of the main circular scale has been crossed in going from one position to other.

Discussions :

1. We must position the source in front of the collimator slit to ensure the image appears bright.

2. We should align the ~~verticle~~ vertical cross-wire, on the center of the cross-wires, with ~~to~~ the same ~~o~~ edge of the slit image. It is ~~imp~~ important to avoid parallax between the cross-wires and the ~~sli~~ slit image.

3. For the final adjustment, we need to rotate the telescope carefully, keeping the tangent screw in the same ~~position~~ direction to prevent backlash error.

4. When measuring the angle of the prism, we must place the vertex of the prism at the center of the table.

5. We should always use ~~the~~ both verniers to avoid eccentric error.

6. When handling a prism, we must avoid touching the refracting faces. We should hold it with the thumb at the bottom and the other fingers at the top. If necessary, we should clean the faces of the prism with a clear piece of fine linen.

7. We should make the width of the slit image as narrow as possible.

8. While taking readings, we must ensure that the zero of the main circle scale is not crossed when moving from one position to another.